Spatial and Cluster Structural Prior-Guided Subspace Clustering for Hyperspectral Image

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***Abstract*— Subspace clustering has achieved remarkable performance for hyperspectral image (HSI). However, existing methods are often computationally expensive and have limited ability to capture the intrinsic structural information of HSI. In this article, we propose a structural prior-guided subspace clustering method, which simultaneously incorporates the local and nonlocal spatial information and the cluster prior information. Accordingly, three efficient regularizations are developed. Considering the local connectivity of pixels, we propose an $\ell_{2,1}$ norm-based constraint on the representation difference matrix to improve the homogeneity of clustering result. Next, to capture the nonlocal geometric structure of HSI, we propose a manifold-based regularization with an adaptively learned landmark graph. Furthermore, we explore the block-diagonal cluster structure of HSI and develop a landmark-based clustering constraint, which makes the representations more favorable for clustering. Our local constraint is imposed on all the data points due to its efficiency and the latter two are solely imposed on landmarks, leading to computationally efficient regularizations. Due to the local constraint, the manifold and cluster structure of the landmarks can be effectively propagated to all the data points. To make our model scalable to large-scale data, we learn a compact dictionary with an orthogonal constraint, significantly reducing the number of parameters. In addition, we propose a novel landmark selection method to support our landmark-based constraints using multiscale super-pixel segmentation and clustering, which improves the uniformity and diversity of landmarks. We also develop an efficient algorithm to solve the proposed model. Experimental results demonstrate that our model outperforms the state-of-the-art.**

***Index Terms*— Clustering, hyperspectral image (HSI), remote sensing, unsupervised classification.**

I. Introduction

HYPERSPECTRAL image (HSI), which captures hundreds of spectral bands, enables a far better discrimination ability between different land cover than conventional multispectral images. Therefore, it finds a number of applications, including precision agriculture, environmental monitoring [1], military defense [2] and urban planning. HSI classification is a fundamental step of these applications. In general, HSI classification methods can be classified as supervised [3], [4], semi-supervised [5], [6] and unsupervised approaches [7], [8], [9], [10], [11], depending on the use of labeled data. As the acquisition of labeled data is labor-intensive and time-consuming, the labeled data is often scarce, which limits the application of supervised and semi-supervised methods. Clustering, which does not rely on labeled data, is able to partition data points into different groups, attracting an increasing attention in the field.

Existing clustering approaches of HSI can be roughly classified into the conventional model-driven clustering methods and the deep learning-based clustering methods [12]. The model-driven clustering methods include centroid-based [13], [14], density-based [15], [16], [17], probability-based [18], [19], [20], graph-based [10], [21], [22], [23], [24], [25], [26] and subspace clustering methods [27], [28], [29], [30], [31], [32]. These methods obtain the clustering results of HSI by analyzing the statistical property of data or by solving the well-formulated convex/nonconvex optimization problems. Deep learning-based clustering methods include self-representation-based [7], [33], [34], [35], [36], [37], [38], [39], autoencoder (AE)-based [40], [41], [42] and contrastive learning-based methods [43], [44], [45]. Compared with the model-driven methods, the deep clustering methods are often more effective in the clustering of HSI with complex data structure due to their superior nonlinear feature extraction ability with deep neural networks. However, deep learning-based methods are less interpretable, and often require powerful GPUs to cope with the high-computational load, which limits their real applications.

Subspace clustering, as a representative model-driven clustering technique, has achieved the current state-of-the-art performance in the clustering of HSI. It assumes that the high-dimensional data of HSI lies in a union of low-dimensional subspaces. By assigning the data points to different subspaces, clustering results can be obtained. Classical subspace clustering methods include sparse subspace clustering (SSC) [27] and low-rank representation (LRR) [46]. Both methods derive cluster affinity of data points by solving a self-representation-based optimization problem where the input data is utilized as a dictionary. The basic idea of SSC and LRR is that with a sparse or low-rank constraint on

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the representation matrix a data point can be represented by a linear combination of the data points from the same subspace. This facilitates the construction of a better similarity matrix and improves the clustering accuracy in the subsequent spectral clustering. However, only the spectral information of HSI is exploited in SSC and LRR, making them sensitive to impulse noise. Therefore, various extensions of SSC and LRR are proposed by incorporating the spatial information of HSI. Those methods typically impose spatial constraints on the representation matrix to promote smoothing coefficients in the local regions. Representative spatial constraints include the joint sparsity constraint [47], the total variation-based constraints [30], [48] and the filtering-based constraints [28], [49]. In addition, some methods [50], [51] adopt graph Laplacian-based constraints on the representation matrix to preserve the manifold structure of data in the low-dimensional subspaces, yielding improved clustering accuracy over SSC and LRR. Moreover, some efforts [33], [38], [39], [52] have been made to extend the above shallow models to the deep versions using AE. These methods exploit the self-representation property of the data in the latent feature space obtained by the encoder. Since the deep features extracted by AE are often more cluster-separable compared with the raw data, the deep subspace clustering methods often outperform the traditional models. Apart from the aforementioned single-view clustering methods, multiview clustering methods [53], [54], [55] are proposed by making use of the rich information from different data sources and yield improved clustering results.

Despite the considerable success of the aforementioned subspace clustering methods, they still face several challenges. An inherent drawback is the lack of task-oriented constraints. The aforementioned methods typically introduce spatial constraints into the unsupervised data-reconstruction-based feature learning framework. However, the learned feature might not be optimal for the clustering task. The second problem is that the existing clustering methods fail to fully exploit the spatial information of HSI. They focus solely on local or nonlocal information, making it difficult to accurately reveal the underlying cluster structure of HSI. Finally, a notable problem is the excessive computational complexity, which results from the use of self-representation dictionary or complex constraints, making them difficult to be deployed in practice.

To alleviate these problems, we propose a computationally efficient subspace clustering model for HSIs, which leverages different spatial priors and cluster structural prior in a unified framework. The main contributions of this work are summarized as follows.

1) We propose a novel structural prior guided subspace clustering framework for HSIs, which simultaneously takes into account the local and nonlocal spatial information and the underlying cluster structure for the first time. Compared with existing subspace clustering methods, the proposed model is more generalized and flexible, which can unveil more accurately the intrinsic cluster patterns of HSI with low-computational complexity.
2) Within the framework, we propose three effective and computationally efficient constraints to regularize our model to yield more clustering-friendly features, including an $\ell_{2,1}$ norm-based local spatial constraint, a landmark-based adaptive graph nonlocal spatial constraint and a landmark-based block-diagonal cluster constraint. Unlike existing approaches, which impose regularizations on all the data points, we impose constraints in a more efficient and flexible manner. Specifically, the designed local spatial constraint is computationally efficient and thus is imposed on all the data points by enforcing spatially adjacent data to yield similar representations. While the latter two regularizations are solely imposed on representative data points, i.e., landmarks, which facilitate to preservation of the nonlocal geometric structure and a more block-diagonal cluster pattern, respectively with a much lower computational complexity. With the local spatial constraint, the manifold and block-diagonal structures of the landmarks can be effectively propagated to all the data points. It is noted that the landmark clustering naturally brings feature learning and clustering within a unified framework, which facilitates obtaining more clustering-friendly representations. The developed regularizations have a low complexity and thus are more scalable to big data.
3) Unlike the huge dictionary employed in self-representation models, we learn a compact dictionary from the input data with an orthogonal constraint to reduce the number of unknown variables, which significantly reduces the computational complexity of the proposed model. In addition, to obtain representative landmarks while maintaining efficiency for the landmark-based constraints, we propose a novel landmark selection method based on multiscale super-pixel segmentation and multiscale clustering, which is demonstrated to perform better than the commonly used random sampling.
4) We design an efficient algorithm to optimize the proposed model. Extensive experiments on benchmark datasets demonstrate that our model outperforms the state-of-the-art in the field.

The rest of this article is organized as follows. Section II reviews the related subspace clustering methods for HSI. Section III presents the proposed clustering model, develops an efficient algorithm to optimize the resulting model and analyzes the time complexity of the proposed algorithm. In Section IV, experiments and analysis on benchmark datasets are conducted. Finally, Section V concludes the article.

II. Related Works

A. Model-Based Subspace Clustering Methods

Subspace clustering, which reveals the underlying cluster structure of data in the low-dimensional subspaces, has been demonstrated to be effective for the clustering of high-dimensional data. Model-based subspace clustering methods learn cluster structure of data by solving convex/nonconvex optimization problems. A number of model-based subspace clustering methods have been proposed for HSI. SSC [27] and LRR [46] are two pioneering works, learning the subspace affinity within a self-representation-based optimization problem. However, they only use the spectral information of HSI, which makes the models sensitive to spectral noise. Subsequent extensions mainly focus on incorporating spatial information to improve the robustness of model to noise.

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Zhang et al. [28] propose a spatial-spectral SSC model by introducing a reference matrix-guided spatial constraint, where the reference matrix is derived by applying a 2-D mean filter to the intermediately obtained representation matrix. Huang et al. [47] impose a joint constraint on the local neighborhood obtained by super-pixel segmentation to reduce the feature variability within cluster, resulting in a significant accuracy improvement. Zhai et al. [48] propose an ℓ_2 norm-based spatial constraint, which facilitates the neighboring pixels to yield similar subspace representations. Other spatial-spectral subspace clustering methods include [50], [51], which employ graph Laplacian constraint to preserve the manifold structure in the low-dimensional subspaces. To reduce the computational complexity of the self-representation-based clustering models, several attempts have been made, including the object-based [56], [57], [58] and the dictionary learning-based methods [59], [60]. The former approach reduces the computational complexity by performing clustering at the object level and the latter one builds a more compact dictionary to reduce the number of parameters to be optimized. Compared with the conventional self-representation models, the two types of methods yield significant performance improvements in terms of running time. Model-based subspace clustering methods also include the kernel-based [61], semi-supervised [62] and multiview clustering approaches [53], [54], [55], [63], [64]. For example, Cai et al. [61] adopt a kernel trick to improve the nonlinear feature extraction ability. Yang et al. [62] incorporate the supervised information from a few labeled training data to construct the similarity matrix, resulting in an improved clustering accuracy. The works in [53], [54], [55], [63], and [64] fuse the rich information from different data sources or feature spaces, leading to better clustering results compared with the single-view clustering models.

B. Deep Subspace Clustering Methods

Powered by the excellent deep feature extraction ability with deep neural network, a number of deep subspace clustering methods are proposed for HSI. These methods typically follow an AE-based data reconstruction framework [33], where a self-representation layer is added to the latent feature space with the aim to learn a better data affinity matrix. For instance, Cai et al. [52] propose a hypergraph regularized deep subspace clustering method for the clustering of HSI, which imposes the hypergraph Laplacian constraint in the latent feature space to yield a better feature representation. A similar graph constraint was imposed on the subspace representation matrix in [7], leading to an improved clustering accuracy. Li et al. [35] embed constrastive learning and self-representation of data into AE and propose a mutual information subspace clustering network. Cai et al. [39] propose a neighborhood constrastive subspace clustering network, which utilizes a super-pixel pooling AE to reduce the computational complexity and a constractive regularization to improve the feature consistency between positive samples. Unlike the aforementioned deep subspace clustering methods, which typically use a convolution AE for feature learning, the work in [65] adopts a multigraph AE to capture the high-order correlations of pixels, yielding improved clustering results. Recently, a self-supervised subspace clustering approach [66] is proposed for HSI, which adopts high-quality pseudo labels to guide the unsupervised feature learning, leading to a significant accuracy improvement. Han et al. [67] develop a deep low-rank graph convolutional subspace clustering method for HSI, which extracts the spatial–spectral information of HSI by combing 1-D and 2-D AEs. In [68], a diffusion subspace clustering model is proposed by incorporating the distribution information and the spatail-spectral information in a unified framework, yielding a notable accuracy improvement. Although deep clustering methods often yield improved clustering accuracy, their computational complexities are often significantly increased. The reason can be caused by the employed self-representation layer with $(MN)^2$ (MN is the number of pixels of an HSI) unknown parameters [33], [38], [52], or by the computationally expensive constraints [52], or by the complex neural network architectures for deep features extraction [68], [69].

III. Proposed Model

In this section, we elaborate our structural prior information guided subspace clustering (SPGSC) model for HSI. The overview of the proposed model is shown in Fig. 1.

A. Problem Formulation

Let $\mathbf{X} \in \mathbf{R}^{B\times MN}$ represent a flattened HSI, where MN is the number of pixels and B is the number of bands. Self-representation-based subspace clustering methods employ the whole input data $\mathbf{X}$ as a dictionary and reveal the underlying cluster affinity by solving the following optimization problem:

$$\arg\min_{\mathbf{C}} \frac{1}{2}\|\mathbf{X}-\mathbf{X}\mathbf{C}\|_F^2 + \alpha\Phi(\mathbf{C}) \tag{1}$$

where $\mathbf{C} \in \mathbb{R}^{MN\times MN}$ is the subspace representation matrix of $\mathbf{X}$, $\Phi(\mathbf{C})$ is a constraint for $\mathbf{C}$ and α is a nonnegative parameter. Typical constraints include $\|\mathbf{C}\|_1$ [27] and $\|\mathbf{C}\|_*$ [46] to promote the sparsity and the low-rankness of $\mathbf{C}$, respectively, where $\|\mathbf{C}\|_1 = \sum|C_{ij}|$ and $\|\mathbf{C}\|_*$ is the sum of the singular values of $\mathbf{C}$. To incorporate the spatial prior information of HSI, different spatial constraints are imposed on the matrix $\mathbf{C}$, including the ℓ_2 norm [48], the $\ell_{1,2}$ norm [32] and the centralized smoothing-based regularizations [49]. These constraints promote a more ideal affinity matrix where the data points from the same subspace are connected. After obtaining $\mathbf{C}$, a similarity matrix is built by $(|\mathbf{C}|+|\mathbf{C}^T|)/2$ and is further fed into spectral clustering to yield the final clustering results. Recent deep learning models [33], [38], [39], [52] extend the model (1) to the deep versions by replacing the raw data $\mathbf{X}$ with deep features extracted by AE, which often obtain improved clustering results.

However, the performance of the aforementioned methods is limited by their excessive computational complexity and insufficient exploitation of the structural prior information of HSI. To alleviate these problems, we propose the following subspace clustering framework:

$$\begin{aligned}&\arg\min_{\mathbf{U},\mathbf{V}} \frac{1}{2}\|\mathbf{X}-\mathbf{U}\mathbf{V}\|_F^2 + \lambda\Gamma(\mathbf{V}) + \beta\Psi(\mathbf{V}) + \gamma\Xi(\mathbf{V})\\ &\text{s.t. } \mathbf{U}^T\mathbf{U} = \mathbf{I}\end{aligned} \tag{2}$$

where $\Gamma(\mathbf{V})$, $\Psi(\mathbf{V})$, and $\Xi(\mathbf{V})$ represent local spatial constraint, nonlocal spatial constraint, and clustering related

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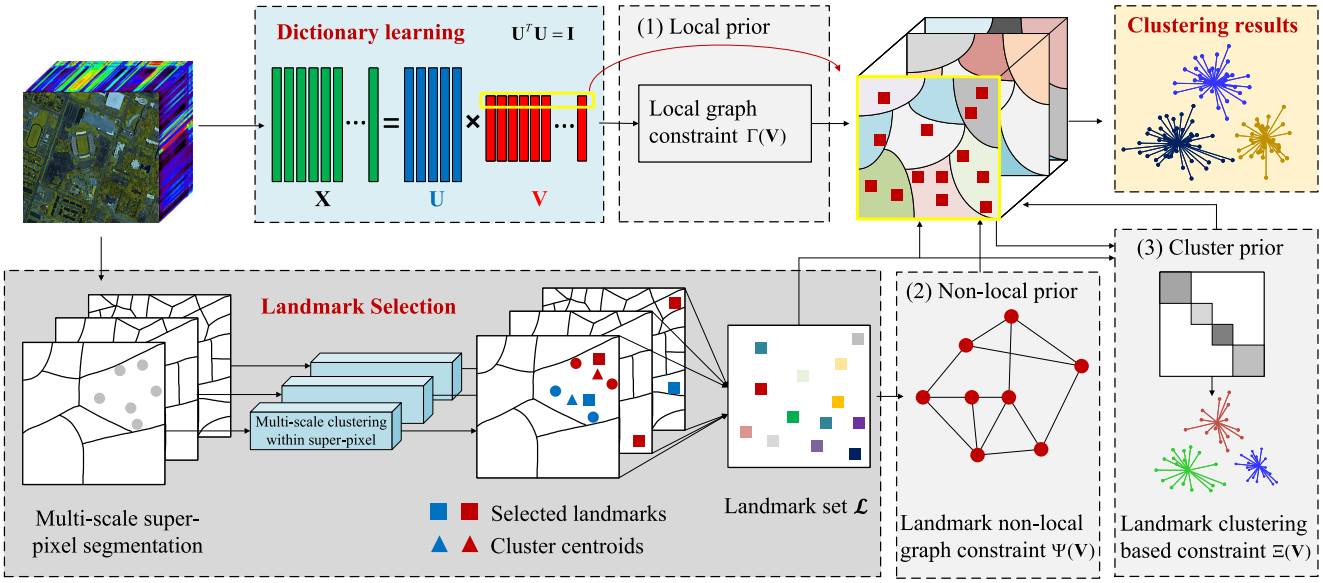


Fig. 1. Framework of the proposed clustering model. The orthogonal dictionary learning is mainly utilized to reduce the overall computational complexity. We incorporate the local, nonlocal and cluster prior information of HSI at the same time by designing local smoothing constraint $\Gamma(\mathbf{V})$, scalable landmark-based graph constraint $\Psi(\mathbf{V})$ and scalable landmark clustering-based constraint $\Xi(\mathbf{V})$, which allows us to learn more accurately the intrinsic cluster structure of data in the low-dimensional subspaces. To preserve well the data structure while maintaining efficiency, we develop an effective landmark selection strategy based on multiscale super-pixel segmentation.

regularization according to the spatial and cluster structural prior information of HSI; λ, β, and γ are three parameters to balance different terms. Unlike the self-representation-based models, which adopt the input data as the dictionary, in (2) we adaptively learn a compact dictionary $\mathbf{U} \in \mathbb{R}^{B\times d}$ from the input data $\mathbf{X}$ with an orthogonal constraint. The orthogonal constraint is essential here. On the one hand, it facilitates a much smaller dictionary compared with the self-representation dictionary, which significantly reduces the number of parameters to be optimized in $\mathbf{V}$, making the proposed model computationally efficient. On the other hand, the orthogonal constraint is able to improve the separability of the features $\mathbf{V}$, which allows for a better clustering result. Next, we detail the different constraints.

B. Local Spatial Constraint

The clustering of HSI aims to assign pixels into different groups. It is known that neighboring pixels of HSI belong to the same class with a high probability. Here, we incorporate such local spatial prior information to formulate our local spatial constraint. Let $\mathbf{x}_i$ and $\mathbf{x}_j$ be the neighboring pixels. Ideally, similar data points should be close in the low-dimensional subspaces. As neighboring pixels are typically similar, the subspace representations of $\mathbf{x}_i$ and $\mathbf{x}_j$, i.e., $\mathbf{v}_i$ and $\mathbf{v}_j$, should be similar except when $\mathbf{x}_i$ and $\mathbf{x}_j$ are located in the boundary regions. Thus, after reshaping each row of the 2-D matrix $\mathbf{V}$ into a matrix and deriving a 3-D tensor as shown at the top of Fig. 1, each slice of the tensor ideally shows a similar spatial structure to that of the original HSI. It is observed that the HSI is typically piecewise smooth due to the homogeneously distributed land cover. To preserve such local structure, we adopt the following $\ell_{2,1}$ norm-based local spatial constraint:

$$\Gamma(\mathbf{V}) = \sum_{i=1}^{MN} \sum_{j\in\mathcal{N}_i} \|\mathbf{v}_i - \mathbf{v}_j\|_2 \tag{3}$$

where $\mathcal{N}_i$ denotes the four adjacent spatial neighbors of the ith pixel in horizontal and vertical directions and $\|\mathbf{x}\|_2 = (\sum_i x_i^2)^{1/2}$. Let $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_{4MN}] \in \mathcal{R}^{d\times 4MN}$ represent the collection of the difference vectors $\{\mathbf{v}_i - \mathbf{v}_j | j \in \mathcal{N}_i\}_{i=1}^{MN}$, where each column of $\mathbf{P}$ is a difference vector, we can reformulate (3) as $\Gamma = \|\mathbf{P}\|_{2,1}$, where $\|\cdot\|_{2,1}$ is an $\ell_{2,1}$ norm defined by $\|\mathbf{X}\|_{2,1} = \sum_j (\sum_i X_{ij}^2)^{1/2}$. As minimizing the $\ell_{2,1}$ norm leads to a column sparsity pattern, most columns of the matrix $\mathbf{P}$ will be $\mathbf{0}$. Thus, most of the adjacent spatial neighbors $\{\mathbf{v}_j\}_{j\in\mathcal{N}_i}$ are close to the centroid $\mathbf{v}_i$, promoting thereby the reduction of within-cluster variance in the low-dimensional subspaces.

Remark 1: The developed local spatial constraint in (3) equals to a graph-based constraint, i.e., $\Gamma(\mathbf{V}) = \sum_{i,j} W_{ij}^l \|\mathbf{v}_i - \mathbf{v}_j\|_2$ given the following similarity matrix:

$$W_{ij}^l = \begin{cases} 1, & j \in \mathcal{N}_i \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Note that although the similarity matrix $\mathbf{W}^l$ is typically huge, most of its elements are zero due to the assumed four-connected neighbors. By assuming a periodic boundary condition, the related minimization problem with the local spatial constraint (3) can be efficiently solved by Fourier transform, which will be shown in Section III-E.

C. Adaptive Landmark-Based Graph Laplacian

The piecewise smoothing structure of land cover can be effectively captured by the local spatial constraint $\Gamma(\mathbf{V})$. It is observed that the same land cover can be distributed nonlocally, leading to a nonlocal spatial structure of HSI. However, relying solely on the local spatial constraint $\Gamma(\mathbf{V})$ fails to capture such nonlocal structure data property. Thus, we propose an adaptive landmark-based graph Laplacian constraint based on the nonlocal spatial structural prior information. Traditional graph Laplacian can be applied as a nonlocal spatial constraint

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to preserve a similar manifold structure to the original data space. However, it requires to build pairwise constraints with a similarity matrix in the transformed domain for all the input data points. As the similarity matrix is typically huge, the resulting optimization problem is computationally expensive, making the traditional graph Laplacian infeasible in practice. Another drawback of the traditional graph Laplacian is that the similarity matrix is often precomputed in the raw data space, which can be easily affected by various degradations of HSI.

To address these problems, we first select representative data points as landmarks to approximate the whole HSI and then build an adaptive graph in the low-dimensional subspaces, which is further employed to build pairwise constraints for the representations of the landmarks. The proposed adaptive landmark-based graph Laplacian constraint is shown as follows:

$$\Psi(\mathbf{V}) = \frac{1}{2}\sum_{i,j\in\mathcal{L}} W_{ij}\|\mathbf{v}_i - \mathbf{v}_j\|_2^2 \tag{5}$$

where $\mathcal{L}$ is a set that contains the index of the selected n ($n \ll MN$) landmarks and $\mathbf{W} \in \mathbb{R}^{n\times n}$ is an iteratively updated similarity matrix.

1) Automatic Landmark Selection: The selection of landmarks is important for the proposed regularization (5), which should preserve a similar data distribution to the original HSI while keeping redundancy as low as possible. Although random sampling is efficient for landmark generation, it neglects the data correlation in the spatial domain, making the selected landmarks highly redundant. Furthermore, how to decide on the number of landmarks is not trivial. Here, we propose an effective landmark selection strategy based on multiscale super-pixel segmentation.

We first segment an HSI into a sequence of segmentation maps $\{\mathcal{S}_i\}_{i=1}^t$, where t is the number of scales and $\mathcal{S}_i = [\mathcal{S}_i^1, \mathcal{S}_i^2, \ldots, \mathcal{S}_i^{N_i}]$ is the ith scale segmented map with $\mathcal{S}_i^j$ being the jth super-pixel and N_i the number of super-pixels in the ith scale. Each scale super-pixel segmentation map is obtained by applying the entropy rate segmentation method [70] in the first principle component of HSI. The number of super-pixels determines the segmentation scale of HSI. A larger value results in a finer segmentation. Following [55], we set the number of super-pixels at the ith scale by $N_i = 2^{i-1}T$, where T is estimated by the gradient information of the first PC of $\mathbf{X}$.

Let $\mathbf{X}_1$ be the first PC of $\mathbf{X}$ and we calculate its gradient matrix $\mathcal{D}(\mathbf{X}_1)$ in horizontal and vertical directions by

$$\mathcal{D}(\mathbf{X}_1)_{ij} = |\mathbf{X}_{1_{i+1,j}} - \mathbf{X}_{1_{i,j}}| + |\mathbf{X}_{1_{i,j+1}} - \mathbf{X}_{1_{i,j}}|. \tag{6}$$

It is reasonable that the image with more edges should be finer segmented. Thus, T is defined by

$$T = \left\lfloor \frac{1}{2\sqrt{MN}}\sum_{ij}(\mathcal{D}(\mathbf{X}_1)_{ij} - \delta)_+ \right\rfloor \tag{7}$$

where $(x)_+ = \max(x/|x|, 0)$, $\lfloor\cdot\rfloor$ denotes the rounding floor operation and $\delta = 1/(MN)\sum_{ij}\mathcal{D}(\mathbf{X}_1)_{ij}$ is a threshold to remove the less significant magnitude of $\mathcal{D}(\mathbf{X}_1)$.

Then, we can obtain the segmentation map $\mathcal{S}_i$ at scale i by setting the number of super-pixel as N_i. To obtain landmarks uniformly, we apply k-means clustering algorithm within each super-pixel and extract the pixels that are closest to the centroids as landmarks. Thus, we obtain the landmark index set of the jth super-pixel at the scale i by

$$\mathcal{L}_i^j = \left\{ p | p = \arg\min_{i\in\mathcal{S}_i^j} \left\| \mathbf{x}_i - \mathbf{c}_k^{ij} \right\|_2^2 \right\} \tag{8}$$

where $\mathbf{c}_k^{ij}$ is the kth cluster centroid after applying k-means in the jth super-pixel at scale i. Then, the ith scale landmark index set are obtained by

$$\mathcal{L}_i = \mathcal{L}_i^1 \cup \mathcal{L}_i^2 \cup \cdots \cup \mathcal{L}_i^{N_i}. \tag{9}$$

By collecting all the landmarks from the t scales, we obtain the landmark set $\mathcal{L}$ as follows:

$$\mathcal{L} = \mathcal{L}_1 \cup \cdots \cup \mathcal{L}_t. \tag{10}$$

Note that if the same landmarks are selected from different scales, only one will be kept in (10).

2) Adaptive Nonlocal Graph Construction: After obtaining the landmark set $\mathcal{L}$, we impose pairwise constraints on the selected landmarks with an adaptive nonlocal graph. Different from the traditional graph construction methods, we build the graph in the low-dimensional subspaces, which is iteratively updated in the optimization procedure. Let $\mathbf{V}_l = \{\mathbf{v}_i\}_{i\in\mathcal{L}}$ represent the landmarks in the low-dimensional subspaces. We first search the K-nearest neighbors of all the landmarks in the Euclidean distance and then build a similarity matrix by

$$\mathbf{W}_{ij} = \begin{cases} e^{\frac{-\|\mathbf{v}_i^l - \mathbf{v}_j^l\|_2^2}{2\sigma^2}}, & \mathbf{v}_i^l \in \mathcal{N}_K(\mathbf{v}_j^l) \text{ or } \mathbf{v}_j^l \in \mathcal{N}_K(\mathbf{v}_i^l) \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where $\mathcal{N}_K(\mathbf{v}_i^l)$ represents the K nearest neighbors of the ith landmark $\mathbf{v}_i^l$. Note that here we remove all iterations of $\mathbf{W}$ and $\mathbf{V}$ for simplicity.

The multiscale landmark selection strategy makes the selected landmarks approximately represent the whole HSI. Combined with the local spatial constraint, the nonlocal regularization of the landmarks indirectly constrains all the data points to preserve a similar nonlocal topological structure of the landmarks in the low-dimensional subspaces.

D. Landmark-Based Clustering Constraint

Most subspace clustering methods perform clustering and subspace representation learning independently. They require a post-clustering method such as spectral clustering to obtain the final clustering results. Such a cascaded clustering framework is prone to produce a suboptimal clustering result as the obtained subspace representation features might not be optimal for the post-clustering algorithm. To improve the feature learning for HSI clustering, we exploit a block-diagonal cluster prior information and propose a landmark clustering-based constraint.

The constraint is inspired by the fact that in the ideal case the similarity matrix of data points should be block-diagonal as only the data points belonging to the same cluster are connected. To preserve such a cluster structure in the low-dimensional subspace, we propose the following constraint:

$$\Xi(\mathbf{V}) = \sum_{i=n-c+1}^{c} \lambda_i(\mathbf{L}) \tag{12}$$

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where $\mathbf{L} = \text{Diag}(\mathbf{S}\mathbf{1}_d) - \mathbf{S}$ is the Laplacian matrix of the similarity matrix $\mathbf{S} \in \mathbb{R}^{n\times n}$ of the landmarks $\mathbf{V}_l$, $\mathbf{1}_d$ is a d-dimensional vector of all ones, c is the number of clusters, and $\lambda_i(\mathbf{L})$ is the ith eigenvalue of $\mathbf{L}$ in decreasing order. We construct the similarity matrix $\mathbf{S}$ by $\mathbf{S} = \mathbf{V}_l^T\mathbf{V}_l$. Note that the constraint of (12) is imposed only on the landmarks, which significantly reduces the computational complexity.

According to [71], the sum of the eigenvalues is equivalent to solving the problem as follows:

$$\sum_{i=n-c+1}^{c} \lambda_i(\mathbf{L}) = \min_{\mathbf{W}_s} \langle \mathbf{L}, \mathbf{W}_s \rangle \quad \text{s.t. } 0 \preceq \mathbf{W}_s \preceq \mathbf{I}, \quad \text{Tr}(\mathbf{W}_s) = c \tag{13}$$

where $\text{Tr}(\cdot)$ is a trace operator that sums the diagonal elements of a square matrix.

Let $\mathbf{W}_s = \mathbf{F}\mathbf{F}^T$ with $\mathbf{F} \in \mathbb{R}^{n\times c}$ and $\mathbf{F}^T\mathbf{F} = \mathbf{I}$. $\mathbf{F}$ can be seen as a cluster indicator matrix. It is easy to check that $\mathbf{W}_s = \mathbf{F}\mathbf{F}^T$ is a feasible solution of (13). Then, the optimization problem of (13) is reformulated as

$$\begin{aligned}\Xi(\mathbf{V}) &= \sum_{i=n-c+1}^{c} \lambda_i(\mathbf{L}) \\ &= \min_{\mathbf{F}^T\mathbf{F}=\mathbf{I}} \text{Tr}(\mathbf{F}^T\mathbf{L}\mathbf{F})\end{aligned} \tag{14}$$

which is exactly the objective function of spectral clustering. Thus, the imposed block-diagonal cluster constraint (12) equals to perform spectral clustering on the landmarks.

Combining the spatial constraints with the landmark clustering-based constraint, we derive the objective function of our clustering model as follows:

$$\begin{aligned}\arg\min_{\mathbf{U},\mathbf{V},\mathbf{F}} &\frac{1}{2}\|\mathbf{X} - \mathbf{U}\mathbf{V}\|_F^2 + \lambda \sum_{i=1}^{MN} \sum_{j\in\mathcal{N}_i} \|\mathbf{v}_i - \mathbf{v}_j\|_2 \\ &+ \frac{\beta}{2} \sum_{i,j\in\mathcal{L}} W_{ij}\|\mathbf{v}_i - \mathbf{v}_j\|_2^2 + \gamma \text{Tr}(\mathbf{F}^T\mathbf{L}\mathbf{F}) \\ \text{s.t. } &\mathbf{U}^T\mathbf{U} = \mathbf{I}, \quad \mathbf{S} = \mathbf{V}_l^T\mathbf{V}_l, \ \mathbf{L} = \text{Diag}(\mathbf{S}\mathbf{1}_d) - \mathbf{S} \\ &\mathbf{F}^T\mathbf{F} = \mathbf{I}.\end{aligned} \tag{15}$$

The proposed model incorporates the local, nonlocal spatial information and the block-diagonal cluster structural prior information at the same time, which enables us to learn the cluster structure of the HSI more accurately. Meanwhile, due to the adopted dictionary learning strategy and landmark-based constraints, the proposed model is more computationally efficient compared with the existing self-representation-based clustering methods.

Remark 2: Existing linear subspace clustering methods, including SSC [27], LRR [46], and elastic net subspace clustering [72], assume that the data points lie on the union of linear subspaces. Despite their theoretically guaranteed performance, they often fail in dealing with the data drawn from nonlinear manifolds [73]. The proposed model can be seen as a nonlinear manifold subspace clustering method. It employs local and nonlocal spatial constraints to preserve the locality and nonlocality of the representations, facilitating to yield similar representations for nearby points, which is essential for clustering tasks. However, such geometric structure preservation of the manifold cannot be achieved by the linear subspace clustering methods, so their performance is often degraded in the HSI clustering as the linearity assumption is often not satisfied due to spectral mixture and noise.

E. Optimization

We use the alternating minimization algorithm to solve the optimization problem in (15), where the three variables $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{F}$ are iteratively updated.

1) Update $\mathbf{V}$*:* The objective function with respect to $\mathbf{V}$ is shown as

$$\begin{aligned}\arg\min_{\mathbf{V}} &\frac{1}{2}\|\mathbf{X} - \mathbf{U}\mathbf{V}\|_F^2 + \lambda \sum_{i=1}^{MN} \sum_{j\in\mathcal{N}_i} \|\mathbf{v}_i - \mathbf{v}_j\|_2 \\ &+ \frac{\beta}{2} \sum_{i,j\in\mathcal{L}} W_{ij}\|\mathbf{v}_i - \mathbf{v}_j\|_2^2 + \gamma \text{Tr}(\mathbf{F}^T\mathbf{L}\mathbf{F}) \\ \text{s.t. } &\mathbf{S} = \mathbf{V}_l^T\mathbf{V}_l, \quad \mathbf{L} = \text{Diag}(\mathbf{S}\mathbf{1}_d) - \mathbf{S}.\end{aligned} \tag{16}$$

As we have

$$\begin{aligned}\text{Tr}(\mathbf{F}^T\mathbf{L}\mathbf{F}) &= \frac{1}{2}\sum_{i,j} S_{ij}\|\mathbf{f}^i - \mathbf{f}^j\|_2^2 \\ &= \frac{1}{2}\sum_{i,j} S_{ij}E_{ij} \\ &= \frac{1}{2}\text{Tr}(\mathbf{E}\mathbf{S}^T) \\ &= \frac{1}{2}\text{Tr}\left(\mathbf{V}_l\mathbf{E}\mathbf{V}_l^T\right)\end{aligned} \tag{17}$$

where $\mathbf{f}^i$ is the ith row of $\mathbf{F}$ and $E_{ij} = \|\mathbf{f}^i - \mathbf{f}^j\|_2^2$, to solve (16), we reformulate (16) as the following problem:

$$\begin{aligned}\arg\min_{\mathbf{V}} &\frac{1}{2}\|\mathbf{X} - \mathbf{U}\mathbf{V}\|_F^2 + \lambda\|\mathbf{H}\mathbf{V}^T\|_{1,2} \\ &+ \beta\text{Tr}\left(\mathbf{V}_l\mathbf{L}_w\mathbf{V}_l^T\right) + \frac{\gamma}{2}\text{Tr}\left(\mathbf{V}_l\mathbf{E}\mathbf{V}_l^T\right) \\ = &\frac{1}{2}\|\mathbf{X} - \mathbf{U}\mathbf{V}\|_F^2 + \lambda\|\mathbf{H}\mathbf{V}^T\|_{1,2} + \text{Tr}\left(\mathbf{V}_l\mathbf{G}\mathbf{V}_l^T\right)\end{aligned} \tag{18}$$

where $\mathbf{H}$ is a combined total variation operator in the horizontal and vertical directions, $\mathbf{H}\mathbf{V}^T$ is the resulting difference matrix of $\mathbf{V}$, and $\mathbf{L}_w = \text{Diag}(\mathbf{W}\mathbf{1}) - \mathbf{W}$ is the Laplacian matrix of $\mathbf{W}$ and $\mathbf{G} = \beta\mathbf{L}_w + (\gamma/2)\mathbf{E}$.

We use alternating direction method of multipliers (ADMM) to solve (18). By introducing auxiliary variables $\mathbf{A} = \mathbf{V}$, $\mathbf{B} = \mathbf{H}\mathbf{V}^T$ and $\mathbf{C} = \mathbf{V}_l$, we have the following equivalent optimization problem of (18):

$$\begin{aligned}\arg\min_{\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{V}} &\frac{1}{2}\|\mathbf{X} - \mathbf{U}\mathbf{A}\|_F^2 + \lambda\|\mathbf{B}\|_{1,2} + \text{Tr}(\mathbf{C}\mathbf{G}\mathbf{C}^T) \\ \text{s.t. } &\mathbf{A} = \mathbf{V}, \quad \mathbf{B} = \mathbf{H}\mathbf{V}^T, \ \mathbf{C} = \mathbf{V}_l.\end{aligned} \tag{19}$$

We obtain the augmented Lagrangian function of (19) by

$$\begin{aligned}L(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{V}) = &\frac{1}{2}\|\mathbf{X} - \mathbf{U}\mathbf{A}\|_F^2 + \lambda\|\mathbf{B}\|_{1,2} + \text{Tr}(\mathbf{C}\mathbf{G}\mathbf{C}^T) \\ &+ \langle \mathbf{Y}_1, \mathbf{A} - \mathbf{V}\rangle + \langle \mathbf{Y}_2, \mathbf{H}\mathbf{V}^T - \mathbf{B}\rangle \\ &+ \langle \mathbf{Y}_3, \mathbf{C} - \mathbf{V}_l\rangle \\ &+ \frac{\mu}{2}\left(\|\mathbf{A} - \mathbf{V}\|_F^2 + \|\mathbf{H}\mathbf{V}^T - \mathbf{B}\|_F^2 \right. \\ &\left. + \|\mathbf{C} - \mathbf{V}_l\|_F^2\right)\end{aligned} \tag{20}$$

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where $\{\mathbf{Y}_i\}_{i=1}^3$ are the Lagrange multipliers and μ is a penalty parameter. We then optimize the problem by updating one variable while fixing others.

The objective function with respect to $\mathbf{A}$ is shown as

$$\arg\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{X}-\mathbf{U}\mathbf{A}\|_F^2 + \frac{\mu}{2}\left\|\mathbf{A}-\mathbf{V}+\frac{\mathbf{Y}_1}{\mu}\right\|_F^2. \tag{21}$$

By setting the derivative of (21) with respect to $\mathbf{A}$ to zero, we update $\mathbf{A}$ as follows:

$$\begin{aligned}\mathbf{A} &= (\mathbf{U}^T\mathbf{U}+\mu\mathbf{I})^{-1}\left(\mu\left(\mathbf{V}-\frac{\mathbf{Y}_1}{\mu}\right)+\mathbf{U}^T\mathbf{X}\right)\\ &= \frac{1}{\mu+1}(\mu\mathbf{V}-\mathbf{Y}_1+\mathbf{U}^T\mathbf{X}).\end{aligned} \tag{22}$$

The objective function with respect to $\mathbf{B}$ is formulated by

$$\arg\min_{\mathbf{B}} \lambda\|\mathbf{B}\|_{1,2} + \frac{\mu}{2}\left\|\mathbf{H}\mathbf{V}^T-\mathbf{B}+\frac{\mathbf{Y}_2}{\mu}\right\|_F^2. \tag{23}$$

Denote by $\mathbf{b}^i$ the ith row of $\mathbf{B}$ and $\mathbf{t}^i$ the ith row of $\mathbf{H}\mathbf{V}^T+\mathbf{Y}_2/\mu$, we solve the problem of (23) in a row-wise manner

$$\arg\min_{\mathbf{b}^i} \lambda\|\mathbf{b}^i\|_1 + \frac{\mu}{2}\|\mathbf{b}^i-\mathbf{t}^i\|_2^2. \tag{24}$$

Then, $\mathbf{b}^i$ is obtained by

$$\mathbf{b}^i = (1-\lambda/\mu/\|\mathbf{t}^i\|_2)_+\mathbf{t}^i \tag{25}$$

where $(x)_+$ is an operator defined as $(x)_+ = \max(x,0)$.

The objective function with respect to $\mathbf{C}$ is shown by

$$\arg\min_{\mathbf{C}} \mathrm{Tr}(\mathbf{C}\mathbf{G}\mathbf{C}^T) + \frac{\mu}{2}\left\|\mathbf{C}-\mathbf{V}_l+\frac{\mathbf{Y}_3}{\mu}\right\|_F^2. \tag{26}$$

By setting the derivative of (26) with respect to $\mathbf{C}$ to zero, we update $\mathbf{C}$ as follows:

$$\mathbf{C} = \mu\left(\mathbf{V}_l-\frac{\mathbf{Y}_3}{\mu}\right)(2\mathbf{G}+\mu\mathbf{I})^{-1}. \tag{27}$$

The objective function with respect to $\mathbf{V}$ is formulated by

$$\begin{aligned}&\arg\min_{\mathbf{V}} \frac{\mu}{2}\left(\left\|\mathbf{A}-\mathbf{V}+\frac{\mathbf{Y}_1}{\mu}\right\|_F^2+\left\|\mathbf{H}\mathbf{V}^T-\mathbf{B}+\frac{\mathbf{Y}_2}{\mu}\right\|_F^2\right.\\ &\qquad\left.+\left\|\mathbf{C}-\mathbf{V}_l+\frac{\mathbf{Y}_3}{\mu}\right\|_F^2\right)\\ &= \frac{\mu}{2}\left(\left\|\mathbf{A}-\mathbf{V}+\frac{\mathbf{Y}_1}{\mu}\right\|_F^2+\left\|\mathbf{H}\mathbf{V}^T-\mathbf{B}+\frac{\mathbf{Y}_2}{\mu}\right\|_F^2\right.\\ &\qquad\left.+\left\|\mathbf{C}'-\mathbf{V}+\frac{\mathbf{Y}_3'}{\mu}\right\|_F^2\right)\end{aligned} \tag{28}$$

where $\mathbf{C}'$ and $\mathbf{Y}_3'$ are expanded matrices of $\mathbf{C}$ and $\mathbf{Y}_3$, respectively by

$$\mathbf{C}'(:,i) = \begin{cases}\mathbf{V}(:,i), & i\notin\mathcal{L}\\ \mathbf{C}(:,i), & i\in\mathcal{L}\end{cases} \tag{29}$$

$$\mathbf{Y}_3'(:,i) = \begin{cases}\mathbf{0}, & i\notin\mathcal{L}\\ \mathbf{Y}_3(:,i), & i\in\mathcal{L}.\end{cases} \tag{30}$$

Algorithm 1 The Algorithm to Update $\mathbf{V}$

Input: $\mathbf{X}$, $\mathbf{U}$, $\mathbf{G}$
Initialize $\{\mathbf{Y}_i\}_{i=1}^3 = \mathbf{0}$, $r=0$, $\varepsilon = 1e-5$, $R=100$
while $||\mathbf{A}-\mathbf{V}||_\infty > \varepsilon$ or $||\mathbf{B}-\mathbf{H}\mathbf{V}^T||_\infty > \varepsilon$ or $||\mathbf{C}-\mathbf{V}_l||_\infty > \varepsilon$ and $r < R$ **do**
  Update $\mathbf{A}$ via (22)
  Update $\mathbf{B}$ via (25)
  Update $\mathbf{C}$ by (27)
  Update $\mathbf{V}$ by (32)
  Update multipliers $\{\mathbf{Y}_i\}_{i=1}^3$ by (33)
  $r := r+1$
end while
Output: $\mathbf{V}$

By setting the derivative of (28) with respect to $\mathbf{V}$ to zero, we update $\mathbf{V}$ by

$$\mathbf{V} = \left(\mathbf{A}+\mathbf{C}'+\frac{\mathbf{Y}_1}{\mu}+\frac{\mathbf{Y}_3'}{\mu}+\left(\mathbf{B}^T-\frac{\mathbf{Y}_2^T}{\mu}\right)\mathbf{H}\right)(\mathbf{H}^T\mathbf{H}+2\mathbf{I})^{-1}. \tag{31}$$

As $\mathbf{H}^T\mathbf{H}+2\mathbf{I}$ is a circulant matrix, we can efficiently update $\mathbf{V}$ with the fast Fourier transform (FFT) as follows:

$$\mathbf{V} = \mathcal{F}^{-1}\left[\frac{\mathcal{F}\left(\mathbf{A}+\mathbf{C}'+\frac{\mathbf{Y}_1}{\mu}+\frac{\mathbf{Y}_3'}{\mu}+\left(\mathbf{B}^T-\frac{\mathbf{Y}_2^T}{\mu}\right)\mathbf{H}\right)}{2+(\mathcal{F}(\mathbf{H}_x))^2+(\mathcal{F}(\mathbf{H}_y))^2}\right] \tag{32}$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the FFT and the inverse FFT, respectively, and $\mathbf{H}_x$ and $\mathbf{H}_y$ are the horizontal and vertical difference operator assuming periodic boundary conditions.

Then, the multipliers $\mathbf{Y}_1$, $\mathbf{Y}_2$ and $\mathbf{Y}_3$ are updated by

$$\begin{cases}\mathbf{Y}_1 = \mathbf{Y}_1+\mu(\mathbf{A}-\mathbf{V})\\ \mathbf{Y}_2 = \mathbf{Y}_2+\mu(\mathbf{H}\mathbf{V}^T-\mathbf{B})\\ \mathbf{Y}_3 = \mathbf{Y}_3+\mu(\mathbf{C}-\mathbf{V}_l).\end{cases} \tag{33}$$

The whole procedure to update $\mathbf{V}$ is shown in Algorithm 1.

2) Update $\mathbf{F}$*:* The objective function with respect to $\mathbf{F}$ is shown as

$$\arg\min_{\mathbf{F}} \mathrm{Tr}(\mathbf{F}^T\mathbf{L}\mathbf{F}), \quad \text{s.t. } \mathbf{F}^T\mathbf{F}=\mathbf{I} \tag{34}$$

which is the standard spectral clustering problem. To solve (34), the c eigenvectors of $\mathbf{L}$ corresponding to the c smallest eigenvalues of $\mathbf{L}$ are computed as the solution of $\mathbf{F}$.

3) Update $\mathbf{U}$*:* The objective function with respect to $\mathbf{U}$ is shown as

$$\arg\min_{\mathbf{U}} \frac{1}{2}\|\mathbf{X}-\mathbf{U}\mathbf{V}\|_F^2, \quad \text{s.t. } \mathbf{U}^T\mathbf{U}=\mathbf{I}. \tag{35}$$

To solve (35), we first reformulate the objective function as

$$\arg\min_{\mathbf{U}^T\mathbf{U}=\mathbf{I}} \frac{1}{2}\mathrm{Tr}((\mathbf{X}-\mathbf{U}\mathbf{V})^T(\mathbf{X}-\mathbf{U}\mathbf{V})) \Longleftrightarrow \arg\max_{\mathbf{U}^T\mathbf{U}=\mathbf{I}} \mathrm{Tr}(\mathbf{U}^T\mathbf{X}\mathbf{V}^T). \tag{36}$$

Let $\mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^T$ be the SVD of $\mathbf{X}\mathbf{V}^T$ and we have the equivalent optimization problem with (36):

$$\arg\max_{\mathbf{U}^T\mathbf{U}=\mathbf{I}} \mathrm{Tr}\left(\mathbf{U}^T\mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^T\right) \Longleftrightarrow \arg\max_{\mathbf{U}^T\mathbf{U}=\mathbf{I}} \mathrm{Tr}\left(\mathbf{V}_1^T\mathbf{U}^T\mathbf{U}_1\mathbf{\Sigma}_1\right). \tag{37}$$

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Algorithm 2 The Proposed Algorithm

Input: $\mathbf{X}$, λ, β, and γ
Initialize $\mathbf{U}$, $\mathbf{F}$, $\varepsilon = 1e-8$, $r = 0$ $R = 100$
while $||\mathbf{V}^{r+1} - \mathbf{V}^r||_F^2 > \varepsilon$ and $r < R$ **do**
    Update $\mathbf{V}^r$ via Algorithm 1
    Update $\mathbf{F}^r$ via solving (34)
    Update $\mathbf{U}^r$ by (38)
    Update $\mathbf{W}$ by (39)
    $r := r + 1$
end while
Output: $\mathbf{V}$

When $\mathbf{U}^T\mathbf{U}_1 = \mathbf{V}_1$, the objective function of (37) has the maximum value. Thus, we update $\mathbf{U}$ by

$$\mathbf{U} = \mathbf{U}_1\mathbf{V}_1^T. \tag{38}$$

Next, we update the similarity matrix $\mathbf{W}$ of the landmarks as follows:

$$\mathbf{W}_{ij} = \begin{cases} e^{\frac{-\|\mathbf{v}_i^l - \mathbf{v}_j^l\|_2^2}{2\sigma^2}}, & \mathbf{v}_i^l \in \mathcal{N}(\mathbf{v}_j^l) \text{ or } \mathbf{v}_j^l \in \mathcal{N}(\mathbf{v}_i^l) \\ 0, & \text{otherwise} \end{cases} \tag{39}$$

where $\mathcal{N}(\mathbf{v}_i^l)$ denotes the K nearest neighbors of $\mathbf{v}_i^l$ in the Euclidean distance.

We iteratively update $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{F}$ until the stop criterion is met. The whole algorithm to solve the proposed model is summarized in Algorithm 2. After matrix $\mathbf{V}$ is obtained, we adopt KNN method to build a similarity matrix, which is further fed to spectral clustering to yield the final clustering results.

Next, we analyze the computational complexity of Algorithm 2. To obtain $\mathbf{V}$ in Algorithm 2, the updates of $\mathbf{A}$, $\mathbf{C}$, and $\mathbf{V}$ in Algorithm 1 have time complexity of $\mathcal{O}(dBMN)$, $\mathcal{O}(n^3 + dn^2)$, and $\mathcal{O}(dMN\log(MN))$, respectively. Since $n, d, B \ll MN$, the time complexity of updating $\mathbf{V}$ is $\mathcal{O}(I_1 dBMN)$, where I_1 is the number of iterations in Algorithm 1. The time complexity of updating $\mathbf{F}$, $\mathbf{U}$, and $\mathbf{W}$ in Algorithm 2 are $\mathcal{O}(cn^2)$, $\mathcal{O}(BdMN)$, and $\mathcal{O}(n(n-1)d)$, respectively. Thus, the time complexity of Algorithm 2 is $\mathcal{O}(I_1 I_2 dBMN)$, where I_2 is the number of iterations in Algorithm 2. Compared with the self-representation-based shallow and deep clustering methods, which have a time complexity of $\mathcal{O}((MN)^3)$, the time complexity of the proposed method is much lower, which is linear with respect to the number of pixels of HSI.

IV. Experiments

A. Datasets

1) Indian Pines: The first test dataset was acquired by the airborne/visible infrared imaging spectrometer (AVIRIS) sensor over the Indian Pines region in Northwestern Indiana. The spatial resolution is 20 m/pixel and the image size is $145 \times 145 \times 200$. In the experiments, due to the effect of water absorption, twenty low signal-to-noise (SNR) spectral bands in 104–108, 150–163, and 200 are removed. Following [78], nine main classes are used for experiments. The false-color image is shown in Fig. 2(a) and the ground truth is shown in Fig. 2(b).

2) HYDICE Urban: The second HSI is the Urban image, which is acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor [79], [80]. The image size is $200 \times 200 \times 210$ with spectrum ranging from 400 to 2500 nm. We remove the severely polluted bands by the atmosphere and water absorption 104–108, 139–151, and 207–210, and keep the remaining 188 bands in the experiments. This dataset contains five classes. The false-color image and ground truth are shown in Fig. 3(a) and (b), respectively.

3) Salinas: The third dataset was collected by the AVIRIS sensor over the Salinas Valley, CA, USA. The test image size is $100 \times 80 \times 204$ after removing water absorption bands. There are six classes in total. We show the false-color image and ground truth of this dataset in Fig. 4(a) and (b), respectively.

B. Experiment Setting

We compare our method with ten benchmark methods, including one classical method, k-means [13], five competing subspace clustering methods, SSC [27], Sketch-SSC [74], JSSC [47], EDSC [76], and Sketch-TV [77], one NMF-based clustering method, HNMF [75], and two deep learning methods, AE [69] and HyperAE [52]. k-means is a classical centroid-based clustering method, which is efficient for clustering large-scale data. SSC employs a self-representation model with a sparsity constraint to learn the affinity matrix of data. JSSC extends SSC to a spatial–spectral subspace clustering model by incorporating spatial information of HSI with a joint sparsity constraint. Sketch-SSC adopts random projection to compress the self-representation dictionary of SSC, resulting in a lower computational complexity. HNMF is a hierarchical clustering method based on nonnegative matrix factorization. AE learns deep features using an AE and inputs the features into k-means. HyperAE applies the self-representation property of data in the latent feature space obtained by an AE, which can be seen as a deep version of SSC. All methods except for AE and HyperAE are implemented in MATLAB. AE and HyperAE are implemented in Python. All codes are run on a computer with an Intel core-i9 12900K CPU. We adopt the commonly used evaluation metrics overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), normalized mutual information (NMI) and adjusted rand index (ARI) for performance evaluation. We denote the running time by T in seconds. Let N represent the number of data points, we compute OA by

$$\text{OA} = \frac{1}{N}\sum_{i=1}^{N}\delta(\text{map}(r_i), l_i) \tag{40}$$

where l_i is the class label of the ith data point, r_i is the clustering label, $\delta(x, y) = 1$ if $x = y$ and is zero otherwise; map$(\cdot)$ is a mapping function used to find the best match between the clustering results and ground truth. Here, the Hungarian algorithm is used for map$(\cdot)$. Let $n_{i,j}$ be the number of samples in class i that are labeled as class j. We compute AA and κ by

$$\text{AA} = \frac{1}{c}\sum_{i}\frac{n_{i,i}}{n_{i,+}} \tag{41}$$

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TABLE I
QUANTITATIVE EVALUATION OF DIFFERENT CLUSTERING METHODS ON THE DATASET INDIAN PINES

No.	k-means [13]	SSC [27]	Sketch-SSC [74]	HNMF [75]	EDSC [76]	JSSC [47]	Sketch-TV [77]	AE [69]	HyperAE [52]	SPGSC
1	29.20	25.63	31.65	28.85	26.33	52.03	**54.27**	27.87	53.57	34.87
2	45.06	43.49	42.17	11.45	38.79	14.46	45.30	0.00	10.72	**61.33**
3	56.11	65.22	63.56	2.48	20.70	65.22	26.09	21.12	0.00	**74.74**
4	81.64	87.95	92.19	81.10	**98.49**	98.02	90.00	41.23	2.33	95.62
5	98.54	75.73	**100**	98.54	53.55	99.58	**100**	94.98	**100**	**100**
6	28.60	37.86	35.60	63.17	35.69	**72.12**	40.43	67.80	5.56	69.55
7	35.80	39.14	37.23	33.44	37.31	41.71	38.74	51.53	**88.11**	72.46
8	22.09	22.60	3.88	26.98	22.09	50.42	30.52	7.25	0.00	**69.65**
9	52.02	46.56	48.77	75.65	97.07	98.42	99.84	61.03	**100**	68.22
OA(%)	44.13	44.38	45.05	44.77	47.60	61.07	56.31	43.25	52.32	**67.96**
AA(%)	49.90	49.35	50.56	46.85	47.78	65.78	58.35	41.42	40.03	**71.83**
κ	0.3595	0.3647	0.3708	0.3664	0.3981	0.5518	0.4984	0.3294	0.4099	**0.6324**
NMI	0.4308	0.4158	0.4529	0.4309	0.4458	0.5536	0.5202	0.4470	0.5350	**0.6390**
ARI	0.2458	0.2311	0.2521	0.2524	0.2934	0.4033	0.3589	0.2479	0.3472	**0.5227**
T (s)	0.4	564	0.6	**0.2**	240	291	0.8	364	188	26

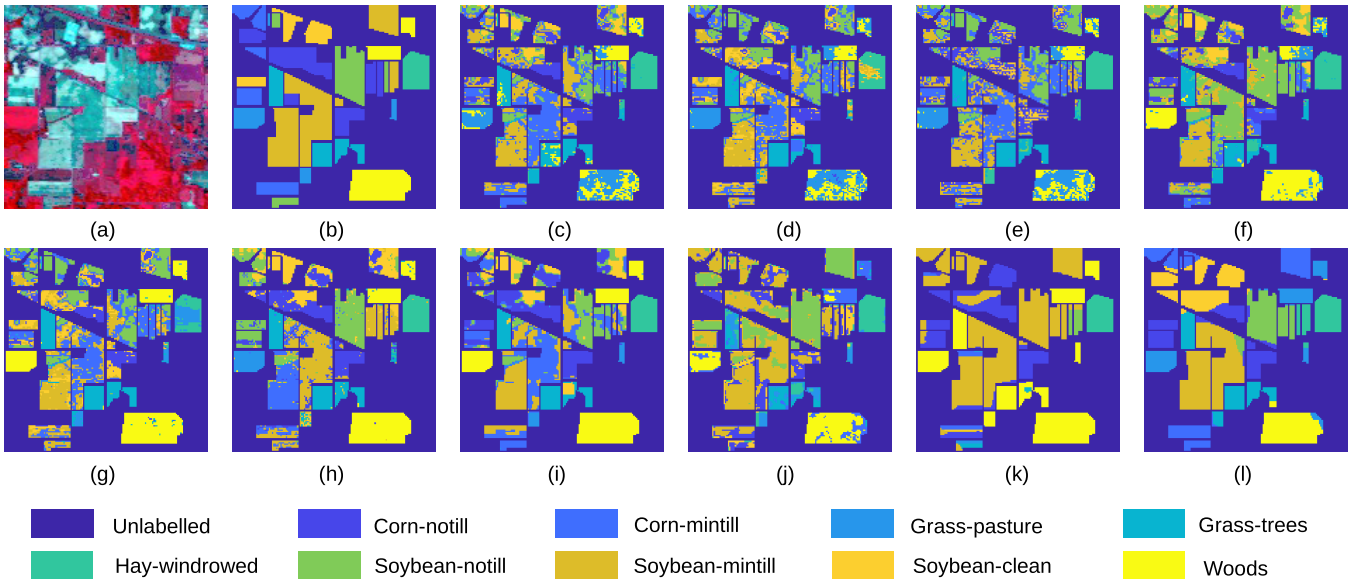


Fig. 2. Indian Pines dataset. (a) False-color image. (b) Ground truth and the clustering results obtained by (c) k-means, (d) SSC, (e) sketch-SSC, (f) HNMF, (g) EDSC, (h) JSSC, (i) sketch-TV, (j) AE, (k) HyperAE, and (l) SPGSC.

$$\kappa = \frac{\frac{1}{N}\sum_i n_{i,i} - \frac{1}{N^2}\sum_i n_{i,+}n_{+,i}}{1 - \frac{1}{N^2}\sum_i n_{i,+}n_{+,i}} \tag{42}$$

where $n_{i,+} = \sum_j n_{i,j}$ is the number of samples in class i and $n_{+,i} = \sum_j n_{j,i}$ is the number of samples that are identified as class i. The NMI score is calculated as

$$\text{NMI} = \frac{\text{I}(l;r)}{\max(\text{H}(l), \text{H}(r))} \tag{43}$$

where $\text{I}(l;r)$ denotes the mutual information between l and r, and $\text{H}(l)$ and $\text{H}(r)$ are their entropies. ARI is compuated by

$$\text{ARI} = \frac{\sum_{ij}\binom{n_{i,j}}{2} - \left(\sum_i \binom{n_{i,+}}{2}\sum_j \binom{n_{+,j}}{2}\right)\Big/\binom{N}{2}}{\frac{1}{2}\left(\sum_i \binom{n_{i,+}}{2} + \sum_j \binom{n_{+,j}}{2}\right) - \left(\sum_i \binom{n_{i,+}}{2}\sum_j \binom{n_{+,j}}{2}\right)\Big/\binom{N}{2}}. \tag{44}$$

A larger value means a better performance. All the methods are fine tuned to yield their optimal results in terms of OA. For the SPGSC method, we fix $K = 50$, $t = 4$, and $\gamma = 10^{-5}$ in all the datasets.

C. Performance Analysis

1) Indian Pines: The clustering results of different methods on the Indian Pines are shown in Table I and Fig. 2. The best result in Table I is annotated in bold and the second-best result is underlined. We set $\lambda = 0.4$, $\beta = 2.5 \times 10^{-3}$ for our method. It is observed that the proposed SPGSC method achieves the best performance in terms of OA, AA, κ, NMI, and ARI. All the subspace-based clustering methods, including SSC, Sketch-SSC, EDSC, JSSC, Sketch-TV, and SPGSC, obtain better results than the centroid-based method k-means, yielding OA improvement by 0.25%, 0.92%, 3.47%, 16.94%, 12.18%, and 23.83%, respectively. Among the subspace-based clustering methods, the spatial–spectral methods, JSSC, Sketch-TV, and SPGSC, outperforms the spectral-based methods, SSC, Sketch-SSC, and EDSC, demonstrating the importance of spatial information. The NMF-based

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TABLE II
QUANTITATIVE EVALUATION OF DIFFERENT CLUSTERING METHODS ON THE DATASET HYDICE URBAN

No.	k-means [13]	SSC [27]	Sketch-SSC [74]	HNMF [75]	EDSC [76]	JSSC [47]	Sketch-TV [77]	AE [69]	HyperAE [52]	SPGSC
1	89.15	59.43	91.16	72.49	78.67	83.89	95.04	92.64	**98.75**	95.20
2	75.96	75.46	73.97	73.68	74.82	97.73	74.82	89.22	**98.30**	76.95
3	59.81	0.31	**95.13**	10.05	42.54	69.23	32.81	2.20	0.16	0.00
4	60.26	69.31	58.11	62.51	58.85	58.38	57.15	58.75	59.10	**73.69**
5	33.55	76.32	20.39	78.95	70.07	49.23	93.31	92.11	**98.03**	94.63
OA(%)	68.92	63.41	69.45	65.33	67.17	71.59	73.02	72.89	76.59	**78.03**
AA(%)	63.75	56.17	67.75	59.54	64.99	**71.69**	70.63	66.98	70.87	68.09
κ	0.5917	0.5249	0.6018	0.5371	0.5674	0.6274	0.6478	0.6472	0.6939	**0.7055**
NMI	0.5831	0.4979	0.6288	0.5169	0.5215	0.5298	0.5846	0.6662	**0.7368**	0.6618
ARI	0.5732	0.4600	0.5913	0.4233	0.4597	0.5128	0.5471	0.5988	**0.6419**	0.6372
T (s)	0.2	694	0.7	**0.1**	165	395	0.9	372	371	41

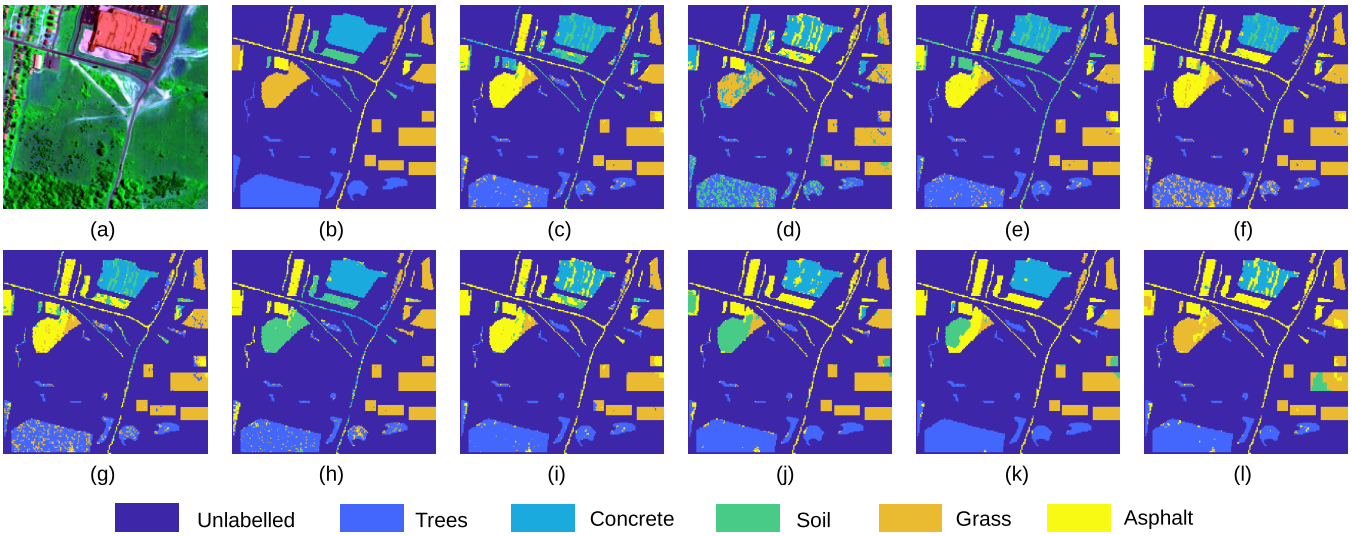


Fig. 3. HYDICE Urban dataset. (a) False-color image. (b) Ground truth and the clustering results obtained by (c) k-means, (d) SSC, (e) sketch-SSC, (f) HNMF, (g) EDSC, (h) JSSC, (i) sketch-TV, (j) AE, (k) HyperAE, and (l) SPGSC.

TABLE III
QUANTITATIVE EVALUATION OF DIFFERENT CLUSTERING METHODS ON THE DATASET SALINAS

No.	k-means [13]	SSC [27]	Sketch-SSC [74]	HNMF [75]	EDSC [76]	JSSC [47]	Sketch-TV [77]	AE [69]	HyperAE [52]	SPGSC
1	**100**	97.24	100	94.77	**100**	97.14	**100**	82.35	99.80	99.90
2	0	**100**	0	**100**	0.00	**100**	0	96.78	**100**	**100**
3	67.33	99.13	99.08	87.07	98.42	96.34	**100**	84.89	**100**	**99.75**
4	93.53	93.25	89.59	0	84.39	97.75	91.56	98.03	**100**	99.02
5	**100**	99.89	**100**	**100**	90.54	**100**	**100**	41.29	93.66	**100**
6	98.69	**100**	**100**	93.01	0.00	**100**	**100**	92.58	**100**	94.32
OA(%)	75.59	98.29	86.48	81.20	79.80	97.87	87.06	80.48	98.89	**99.53**
AA(%)	76.59	98.25	81.45	79.14	62.22	98.54	81.93	82.65	**98.91**	98.83
κ	0.6938	0.9780	0.8243	0.76	0.7363	0.9728	0.8317	0.7557	0.9858	**0.9939**
NMI	0.8250	0.9545	0.8927	0.8387	0.8180	0.9439	0.9081	0.7697	0.9781	**0.9828**
ARI	0.6806	0.9681	0.8538	0.7659	0.7956	0.9498	0.8664	0.7076	0.9869	**0.9903**
T (s)	**0.098**	146	0.176	0.178	31	82	0.3	252	1895	11

clustering method HNMF yields competing performance compared with SSC. Instead of promoting a sparse graph in SSC, EDSC builds a dense graph by imposing Frobenius norm on the subspace representation matrix, yielding an improved clustering result. Sketch-SC and Sketch-TV adopt random projection technique to build a compact dictionary to reduce the computational complexity of SSC. It is shown that both methods perform better than SSC. Compared with Sketch-SC, Sketch-TV incorporates spatial information and yields improved performance. The deep learning-based approach AE does not obtain good result on this dataset. A possible reason is that the latent features learned by AE are not "ball-like" distributed, resulting in degraded performance in the subsequent k-means clustering method. Another deep learning-based approach HyperAE integrates self-representation property of data into AE and adopts a

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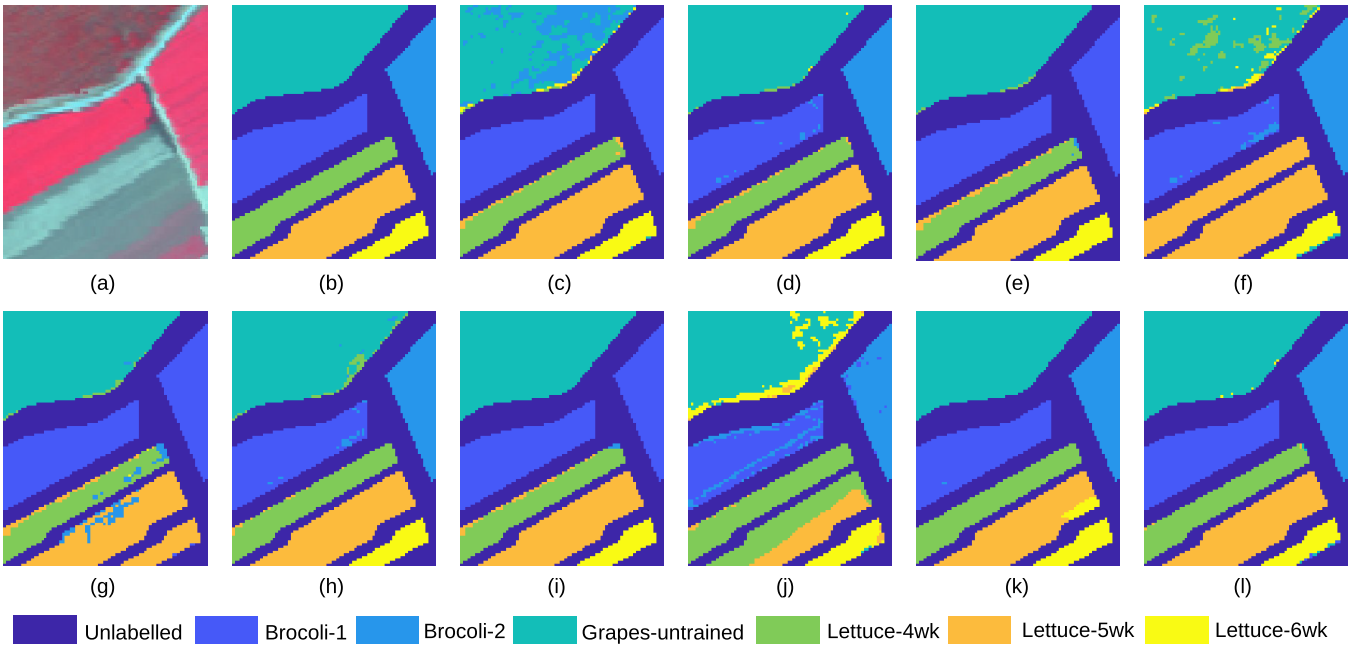


Fig. 4. Salinas dataset. (a) False-color image. (b) Ground truth and the clustering results obtained by (c) k-means, (d) SSC, (e) sketch-SSC, (f) HNMF, (g) EDSC, (h) JSSC, (i) sketch-TV, (j) AE, (k) HyperAE, and (l) SPGSC.

hypergraph-based constraint on the representation matrix. Benefiting from the deep feature and graph constraint, HyperAE performs better than most of the subspace-based methods. However, it is noticed that HyperAE performs worse than the shallow model JSSC, which adopts a joint constraint within an over-segmented super-pixel. This demonstrates that an effective local constraint might be more effective for accuracy improvement compared with the deep feature extraction in AE. SPGSC integrates local and nonlocal constraints within a unified clustering framework, resulting in a better clustering performance. It is observed that SPGSC yields higher accuracies for most classes, especially for the classes "corn-mintill," "grass-pasture," and "soybean-clean" with a significant improvement. The visual results in Fig. 2 indicate that the clustering map of SPGSC is closest to the ground truth. In general, the spatial–spectral clustering methods JSSC, Sketch-TV, Hyper-AE and SPGSC yield better clustering maps than the spectral-based clustering methods k-means, SSC, Sketch-SSC, HNMF, EDSC, and AE. The spectral-based methods are more susceptible to spectral variability and impulse noise, as evidenced by the salt-peper-like wrong clustering results. It is observed that HNMF is the fastest algorithm. Compared with the self-representation models SSC, JSSC, EDSC, and HyperAE, the running time of SPGSC is significantly reduced while achieving remarkable accuracy improvement.

2) HYDICE Urban: We show the clustering results of different methods on the dataset HYDICE Urban in Table II and Fig. 3. We set λ and β for SPGSC as 0.1. The results show that SPGSC outperforms other state-of-the-art clustering methods with the highest OA. In this dataset, k-means performs comparably to the subspace clustering methods Sketch-SSC and EDSC. Again, the spatial–spectral clustering methods JSSC, Sketch-TV, HyperAE, and SPGSC yield improved performance over the spectral-based methods SSC, Sketch-SSC, NHMF, and EDSC. Compared with SSC, its extensions Sketch-SSC, EDSC, JSSC, Sketch-TV, HyperAE, and SPGSC show much better performance, which can be mainly attributed to the elegantly designed dictionaries or spatial constraints. The deep learning-based methods AE and HyperAE perform very well on this dataset, yielding 3.97% and 7.67% OA improvement over k-means. However, compared with the spatial–spectral shallow methods Sketch-TV and SPGSC, AE does not show improved performance, which is consistent with the results in the Indian Pines. The reason is that the deep features extracted by AE might not preserve well the spatial structure of data in the latent space. This verifies again the importance of incorporating spatial prior information in the clustering of HSI. In contrast, HyperAE incorporates spatial constraint in the model and yields improved accuracy than AE. The shallow method SPGSC obtains better performance than the deep learning method HyperAE in terms of OA. However, as the whole third class is wrongly clustered into other clusters, the performance of SPGSC is slightly worse than the HyperAE in terms of other evaluation metrics. For the class-specific accuracies, SPGSC does not show consistently better results over all the classes. This can be caused by the fact that the optimal performance of different methods are determined by the OA, which cannot guarantee improved accuracies for all the classes when the classes in the data are imbalanced. Similar to the results in Indian Pines, HNMF, and k-means are the two most efficient clustering algorithms. The running time of Sketch-SSC, Sketch-TV, and SPGSC is much shorter than the self-representation-based models because of the adopted compact dictionaries.

Regarding to the visual results in Fig. 3, the clustering maps of SPGSC and HyperAE show the best consistency with the ground truth. Compared with others, they contain much less isolated pixels which are wrongly classified. Although other spatial–spectral clustering methods JSSC, Sketch-TV suffer from this problem to a certain degree as shown in Fig. 3(h) and (i), they perform much better than the spectral-based

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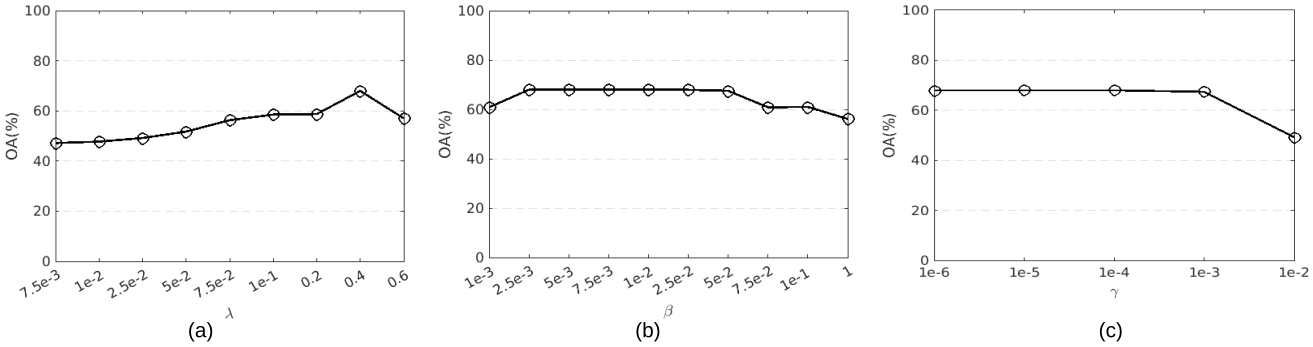


Fig. 5. Influence of parameters (a) λ, (b) β, and (c) γ on the performance of the proposed method in the Indian Pines dataset.

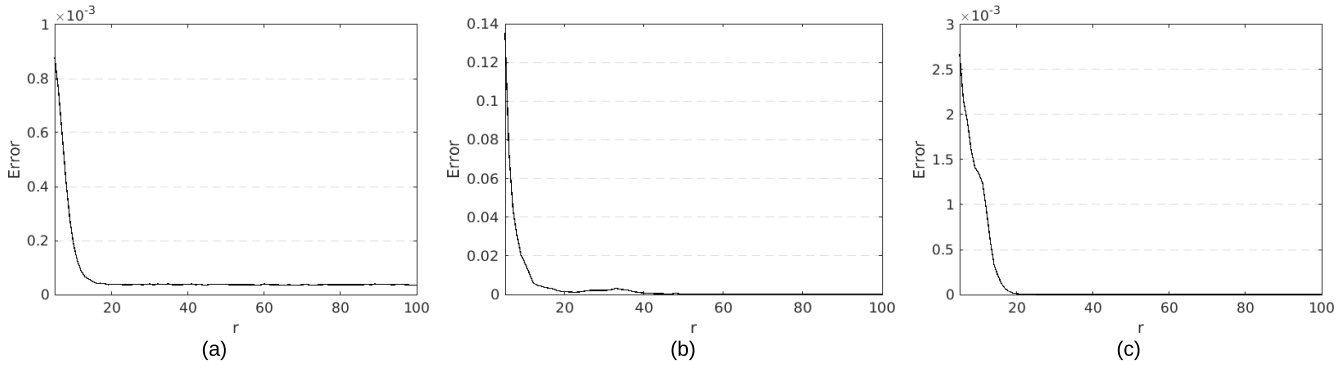


Fig. 6. Convergence property of the proposed method in (a) Indian Pines, (b) HYDICE Urban, and (c) Salinas.

clustering methods k-means, SSC, Sketch-SSC, HNMF, and EDSC. Compared with the clustering map of HyperAE in Fig. 3(k), SPGSC shows a significantly improved result for the class "Grass" as shown in Fig. 3(l).

3) Salinas: The clustering results on Salinas are shown in Table III and Fig. 4. We set λ and β for the SPGSC method as 7.5×10^{-3} and 0.1, respectively. It is observed that SPGSC yields the best performance in terms of OA, κ, NMI, and ARI. HyperAE obtains the second-best result. The centroid-based method k-means does not perform well in the Salinas, while the subspace clustering method SSC yields significant OA improvement over k-means by 22.7%. Other subspace clustering methods, Sketch-SSC and Sketch-TV, also show performance improvement over k-means, but their results get worse compared with SSC. One possible reason can be caused by the undesired dictionary constructed by random projection, which leads to inseparable features in the low-dimensional subspaces. JSSC performs stably on all the three datasets, consistently achieving OA improvements over the spectral-based subspace clustering methods. The deep learning method AE obtains better result than k-means but performs worse than most of the subspace clustering methods. Compared with HyperAE, SPGSC yields better result with 0.64% OA improvement, which verify that model-driven method with appropriate prior information is able to yield competing performance compared with the current data-driven technique. The results show that the running time of k-means, HNMF, Sketch-SSC, and Sketch-T is less than 1 s, which is much shorter than others. Compared with the deep learning method HyperAE, which yields a competing performance in terms of OA, the proposed model obtains an improved performance both in clustering accuracy and computational complexity, which can be mainly attributed to the adaptively learned compact dictionary and the proposed computationally efficient constraints.

Fig. 4 shows that the clustering maps of the spectral-based clustering methods k-means, HNMF, EDSC, and AE are more contaminated by the impulse noise compared with the spatial–spectral clustering methods, which is consistent with the results in Figs. 2 and 3. SPGSC yields the best visual result as shown in Fig. 4(l).

D. Parameter Study

In this section, we investigate the influence of different parameters on the performance of SPGSC on the Indian Pines dataset. The results with respect to λ, β, and γ are shown in Fig. 5. It is observed in Fig. 5(a) that the OA is increased at the beginning with a larger λ and reaches to the best result when λ is set to 0.4. When λ is too large, the performance starts to drop. Fig. 5(b) shows that the performance of SPGSC is more stable with respect to β than λ. More specifically, SPGSC yields saturated results when β is in the range of $[2.5 \times 10^{-3}, 5 \times 10^{-2}]$. Too low or too high β degrades the performance of SPGSC. In Fig. 5(c), the OA of SPGSC is very stable when γ is less than 0.001 but drops dramatically when $\gamma = 0.01$. More experiments on the parameter study, which are conducted on three datasets, are shown in the supplementary material. Based on the results, it is suggested to set $\lambda \in [5 \times 10^{-2}, 1]$, $\beta \in [5 \times 10^{-3}, 5 \times 10^{-2}]$ and $\gamma \in [10^{-6}, 10^{-5}]$ for the proposed model in real application.

E. Convergence Analysis

We show the convergence curves of the proposed algorithm in Fig. 6. Three datasets are tested, including Indian Pines, HYDICE Urban, and Salinas. The x-axis is the number of iterations and the y-axis represents the error $\|\mathbf{V}^{r+1} - \mathbf{V}^{r}\|_F^2$.

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TABLE IV
SCHEMES FOR ABLATION STUDY

Schemes	Local	Nonlocal	Landmark clustering	Landmarks
Scheme-1	✓	✗	✓	✓
Scheme-2	✗	✓	✓	✓
Scheme-3	✓	✓	✗	✓
Scheme-4	✓	✓	✓	✗
SPGSC	✓	✓	✓	✓

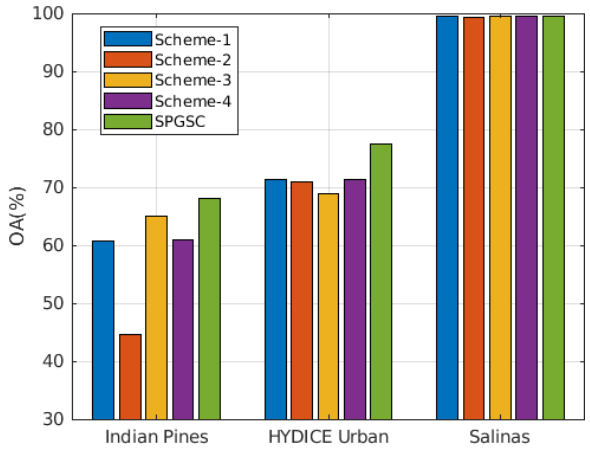


Fig. 7. Effect of different components on the performance of SPGSC in Indian Pines, HYDICE Urban, and Salinas.

The results show that the error drops dramatically in the first few iterations and becomes stable when r is sufficiently large. We find that the solution is often saturated to a stable value within 20 iterations using the proposed algorithm, demonstrating a fast convergence property of Algorithm 2.

F. Ablation Study

We investigate the influence of different components on the performance of SPGSC in this section. Five schemes are designed as shown in Table IV. Specifically, "Scheme-1" is the reduced model of SPGSC by removing the nonlocal constraint; "Scheme-2" is the reduced model of SPGSC by removing the local spatial constraint; "Scheme-3" means that the joint clustering term of SPGSC is removed; and "Scheme-4" replaces the proposed landmark selection in SPGSC with random sampling. We conduct ablation study on three datasets and report the results in Fig. 7. It is observed that compared with SPGSC, the reduced models in different schemes yield degraded performance, which demonstrates the effectiveness of each design. The results show that the influence of different components on the performance of SPGSC varies from dataset to dataset. In Indian Pines, the local spatial constraint makes the largest contribution to SPGSC as evidenced by "Scheme-2" which yields the largest accuracy drop of 23.32% compared with SPGSC. In HYDICE Urban, the joint clustering term is more significant than others, resulting in the largest OA drop of 8.68%. While in Salinas, all schemes perform comparably to SPGSC and "Scheme-2" obtains a slightly worse result than other schemes. The results show that the OA of "Scheme-4" is reduced by 7.04%, 6.14%, and 0.13% compared with SPGSC in Indian Pines, HYDICE Urban, and Salinas, respectively. This verifies that the proposed landmark selection strategy is much more effective than the random sampling technique. Compared with "Scheme-1," SPGSC yields OA improvement of 7.16%, 6.25%, and 0.08% in Indian Pines, HYDICE Urban, and Salinas, respectively.

V. CONCLUSION

In this article, we propose a SPGSC model by incorporating the spatial prior information and the cluster prior information. Local and nonlocal spatial information of HSI are considered at the same time by imposing the developed local $\ell_{2,1}$ norm-based graph constraint and the landmark-based nonlocal graph constraint on the representations of data in the low-dimensional subspaces. The spatial constraints enhance the consistency of features within cluster. To improve the intercluster feature consistency, we propose a landmark clustering-based constraint based on the cluster block-diagonal structural prior information, which naturally brings subspace learning and clustering within a single framework, making the learned features more clustering-friendly. To improve the efficiency of the proposed model, we utilize a dictionary learning strategy with an orthogonal constraint to construct a compact dictionary and propose a novel landmark selection strategy based on multiscale segmentation and clustering. Moreover, we develop an effective algorithm to solve the resulting model based on alternating minimization and ADMM algorithms. Experimental results on real HSIs show that the proposed model yields improved clustering results over the current state-of-the-art.

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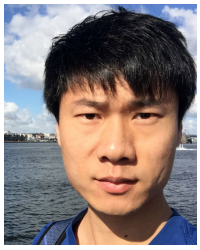
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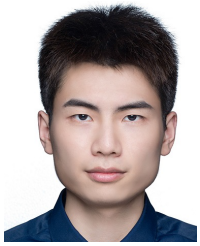
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